

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1	- This section contains FIFTEEN (15) questions. Each question has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers.
SECTION - 2	- This section contains EIGHT (08) questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s). - For each correct answer you will get TWO marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. - There are NO NEGATIVE marks for wrong answers.
SECTION - 3	- This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. - Under each paragraph there will be three questions that has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers.
SECTION - 4	- This section contains THREE (03) questions. These questions are MATCHING TYPE BASED QUESTIONS. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers.

Section - 1

- A student saves money in such a way that his savings form an A.P. If he saves ₹200 in the first month and ₹50 more every month, what will be his savings in the 12th month?
(A) ₹750 (B) ₹800
(C) ₹850 (D) ₹900
- The sum of the roots of a quadratic equation is 6 and their product is 8. If

both roots are increased by 2, what is the new quadratic equation?

- (A) $x^2 - 10x + 20 = 0$
(B) $x^2 - 10x + 24 = 0$
(C) $x^2 - 8x + 12 = 0$
(D) $x^2 - 6x + 8 = 0$

- A drone moves from point A(2, 3) to point B(10, 3) along a straight horizontal path. What is the distance travelled?

- (A) 6 units (B) 7 units
(C) 8 units (D) 10 units

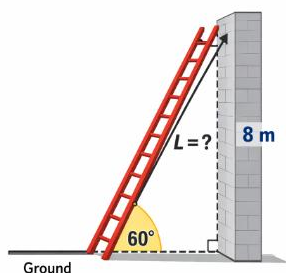
4. A company models its profit (in ₹ thousands) by $P(x) = 2x^3 - 5x^2 + 3x + 7$, where x is units produced (in hundreds). What is the remainder when profit is divided by $x - 2$?

- (A) 5 (B) 7
(C) 9 (D) 11

5. A father is 4 times as old as his son. After 10 years, he will be 2 times as old as his son. Find the present age of the son.

- (A) 10 years (B) 12 years
(C) 8 years (D) 5 years

6. A ladder makes an angle of 60° with the ground and reaches a wall of height 8 m. What is the length of the ladder?



- (A) 16 m (B) $\frac{16}{\sqrt{3}}$ m
(C) $\frac{8}{\sqrt{3}}$ m (D) 4 m

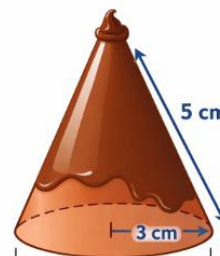
7. In a circle, a chord AB is 10 cm long. The perpendicular from the centre O to AB meets it at M. What can be concluded?

- (A) OM bisects AB
(B) OM is parallel to AB
(C) OM is a tangent
(D) OM is equal to AB

8. A shopkeeper packs 1800 biscuits into identical packets with no biscuits left. Which of the following could be the number of packets?

- (A) 17 (B) 25
(C) 32 (D) 37

9. A cone has radius 3 cm and slant height 5 cm. A chocolate layer is applied only on the curved surface. What area is covered?



- (A) 15π (B) 30π
(C) 45π (D) 60π

10. A surveyor measures the height of a tower using its shadow. At the same time, a 2 m stick casts a shadow of 3 m, while the tower casts a shadow of 45 m. What is the height of the tower?

- (A) 20 m (B) 25 m
(C) 30 m (D) 35 m

11. A school collects marks of students directly by conducting a test for analysis. What type of data is this?

- (A) Secondary data (B) Primary data
(C) Raw data (D) Grouped data

12. Two wires of lengths 306 cm and 657 cm are to be cut into equal pieces of maximum length. What is the length of each piece?

- (A) 3 cm (B) 9 cm
(C) 27 cm (D) 51 cm

13. A bag contains 3 black and 4 red balls. One ball is drawn at random. What is the probability of getting a red ball?

- (A) $\frac{3}{7}$ (B) $\frac{4}{7}$
(C) $\frac{1}{2}$ (D) $\frac{2}{7}$

14. A man standing on a cliff observes a boat at an angle of depression of 45° . If the height of the cliff is 50 m, how far is the boat from the base of the cliff?

(A) 50 m (B) $50\sqrt{2}$ m
(C) 25 m (D) $25\sqrt{2}$ m

15. A rectangular garden area is modeled by polynomial $A(x) = x^2 - 7x + 10$. Which dimensions make the area zero (invalid garden)?

(A) 2, 5 (B) 1, 10
(C) 3, 7 (D) 4, 6

Section - 2

16. A number of trees are planted in rows such that the number of trees in each row forms an A.P. If the first row has 10 trees and the common difference is 2, how many trees are there in the 15th row?

(A) 38 (B) 36
(C) 19×2 (D) 18×2

17. Points A(1, 2), B(4, 6), and C(7, 10) represent locations of sensors. Which statements are correct?

(A) $AB + BC = AC$
(B) Points are collinear
(C) Distance $AB = 5$ units
(D) Distance $BC = 4$ units

18. A bag has 4 green, 3 blue, and 3 red balls. One ball is drawn.

Event E = "not green".

Which are correct?

(A) $P(E) = \frac{6}{10}$
(B) Complement of E is "green"
(C) $P(E) + P(\text{green}) = 1$
(D) $P(E) = \frac{4}{10}$

19. 8 men and 12 boys complete a work in 10 days, while 6 men and 8 boys complete it in 14 days. Based on this situation, which of the following are correct?

(A) One man alone can finish the work in 140 days
(B) One boy alone can finish the work in 280 days
(C) One man is twice as efficient as one boy
(D) One boy is half as efficient as one man

20. A cone and a cylinder have the same base radius and height. Both are used to store water.

Which of the following statements are correct?

(A) Volume of cone is one-third of cylinder
(B) Cylinder volume is three times that of cone
(C) Cone must be filled three times to fill cylinder
(D) Both have equal volume

21. A rectangle has area 48 m^2 and length exceeds breadth by 2 m. Which of the following are correct?

(A) Equation formed is $x(x + 2) = 48$
(B) Equation becomes $x^2 + 2x - 48 = 0$
(C) Length = 8 m
(D) Breadth = 6 m

22. A student writes numbers: 5, 9, 13, 17, ...

Which are correct?

(A) It is an A.P.
(B) Common difference = 4
(C) 10th term = 41
(D) General term = $5 + (n - 1)4$

23. Two linear equations represent the sides of a geometric figure: $5x - 2y = 4$
 $10x - 4y = 8$

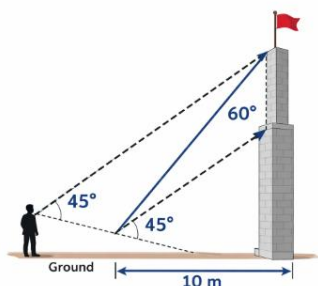
Which of the following statements are correct?

- (A) The system has infinitely many solutions
 (B) The lines represented by the equations coincide
 (C) The equations are dependent
 (D) The system has a unique solution

Section - 3

Passage 1: Building and Flagstaff

A tower stands on a horizontal ground. A flagstaff is fixed on its top. The angle of elevation of the top of the tower from a point on the ground is 45° , and that of the top of the flagstaff is 60° . The distance of the point from the base of the tower is 10 m.



24. What is the height of the tower?
 (A) 10 m (B) $10\sqrt{2}$ m
 (C) 20 m (D) $5\sqrt{2}$ m
25. What is the total height (tower + flagstaff)?
 (A) $10\sqrt{3}$ m (B) $20\sqrt{3}$ m
 (C) $5\sqrt{3}$ m (D) 30 m
26. What is the height of the flagstaff?
 (A) $10(\sqrt{3} - 1)$ m (B) $10(\sqrt{3} + 1)$ m
 (C) $5(\sqrt{3} - 1)$ m (D) $5(\sqrt{3} + 1)$ m

PASSAGE-2

In a cricket match, Virat Kohli took wickets based on a quadratic pattern. The number of wickets taken by Kohli is one less than twice the wickets taken by Jasprit Bumrah. The product of the wickets taken by both players is 15.

27. If Bumrah takes x wickets, which quadratic equation represents the situation?

- (A) $x^2 - 2x - 15 = 0$
 (B) $2x^2 - x - 15 = 0$
 (C) $2x^2 - x + 15 = 0$
 (D) $x^2 + 2x - 15 = 0$

28. Find the number of wickets taken by Bumrah.

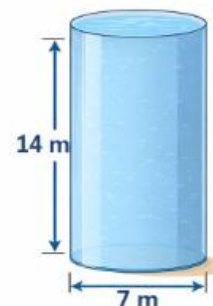
- (A) 3 (B) 5
 (C) -3 (D) 1

29. Find the number of wickets taken by Kohli.

- (A) 5 (B) 9
 (C) 3 (D) 7

PASSAGE-3

A cylindrical tank has radius 7 m and height 14 m. It is completely filled with water. Half of the water is removed. Then the tank is refilled again.



30. What is the volume of the tank?

- (A) 686π (B) 1372π
 (C) 343π (D) 2744π

31. Volume of water removed?

- (A) 686π (B) 343π
(C) 1372π (D) 490π

32. If radius doubles, volume becomes:

- (A) Double (B) Triple
(C) Four times (D) Half

Section - 4

33. A delivery company tracks positions of warehouses and checkpoints on a coordinate plane.

Column I (Situation)	Column II (Result)
(a) Distance between A(2,3) and B(8,3)	(i) (6,8)
(b) Midpoint of A(2,4) and B(8,10)	(ii) 6 units
(c) Centroid of triangle with vertices (0,0), (6,0), (0,6)	(iii) (2,2)
(d) Point dividing A(2,4), B(8,10) in ratio 2:1	(iv) (5,7)

- (A) a-ii, b-i, c-iii, d-iv
(B) a-ii, b-iv, c-iii, d-i
(C) a-i, b-ii, c-iii, d-iv
(D) a-ii, b-i, c-iv, d-iii

34. A student tosses coins multiple times and records different types of events.

Match each situation in Column I with the correct result in Column II.

Column I (Situation)	Column II (Result)
(a) Total outcomes when two coins are tossed	(i) 8

Column I (Situation)	Column II (Result)
(b) Total outcomes when three coins are tossed	(ii) 4
(c) Event of exactly one head in two coin tosses	(iii) 2 outcomes
(d) Event of at least one head in two coin tosses	(iv) 3 outcomes

- (A) a-ii, b-i, c-iii, d-iv
(B) a-ii, b-iv, c-iii, d-i
(C) a-i, b-ii, c-iii, d-iv
(D) a-ii, b-i, c-iv, d-iii

35. A student forms quadratic equations based on given roots and conditions.

Match each condition in Column I with the correct equation in Column II.

Column I (Conditions)	Column II (Equations)
(a) Roots are 2 and 3	(i) $x^2 - 5x + 6 = 0$
(b) Roots are equal and sum = 6	(ii) $x^2 - 6x + 9 = 0$
(c) Product of roots = 8 and sum = 6	(iii) $x^2 - 6x + 8 = 0$
(d) Roots are 4 and -1	(iv) $x^2 - 3x - 4 = 0$

- (A) a-ii, b-i, c-iii, d-iv
(B) a-ii, b-iv, c-iii, d-i
(C) a-i, b-ii, c-iii, d-iv
(D) a-ii, b-i, c-iv, d-iii